

SP-2100 GROUNDWATER WELL

PART 1 GENERAL

1.1 SCOPE OF WORK

- A. The work covered by this section includes furnishing of all plant, labor, tools, equipment and materials, and performing all operations in connection with the drilling, casing, developing, and testing of a private water supply production well and site cleanup, all in accordance with these specifications and applicable drawings. The work also includes, if necessary, well abandonment in accordance with State of Idaho regulations.
- B. Drilling of the groundwater well is to be completed under this this Contract.
- C. The total depth of the production well is estimated at 250 feet. The actual well depth will be determined by the pump test during construction. The well depth and/or production rate will be deemed adequate when the pumping test maintains at least 10 gpm for 24 hours or 5 gpm for 72 hours. If the pump test yields between 5 gpm and 10 gpm, complete pump test and request determination from Contracting Officer before moving to "Priority 2" location.
- D. The total drill depth at one well site shall not exceed 300 feet.
- E. If the "Priority 1" well location does not meet production requirements outlined above, an alternative drilling location – "Priority 2" – has been identified. The Contractor shall notify the Contracting Officer prior to commencing drilling at the "Priority 2" location and prior to demobilizing from site.
- F. The permanent pump and pitless adapter will be installed as part of Phase 2 under a separate Contract.
- G. Contractor shall satisfy himself as to the condition of access to the site for all equipment. No additional payment will be made for access improvements.

1.2 SECTION INCLUDES

- A. Drilling Permit
- B. 12" Drilling
- C. 10" Drilling
- D. 10" Diameter Outer Steel Casing
- E. 6" Diameter Steel Well Casing
- F. 4" Diameter PVC Casing
- G. 4" Diameter PVC Well Screen

- H. 6" Temporary Well Cap
- I. Grouting
- J. Testing for Yield and Drawdown
- K. Chemical Analysis of Water
- L. Bacteriological Test

1.3 RELATED SECTIONS

- A. ISPWC Section 401 – Water Pipe and Fittings
- B. ISPWC Section 402 – Hydraulic Valves

1.4 REFERENCES

- A. ANSI: American National Standards Institute
- B. AISI: American Iron and Steel Institute
- C. IEEE: Institute of Electrical and Electronics Engineers
- D. NEMA: National Electrical Manufacturers Association
- E. AWWA: American Water Works Association
- F. ISO: International Standards Organization

1.5 SUBMITTALS

- A. Submit shop drawings, manufacturer's certifications, or methods of construction for the items shown below:
 - 1. Well casing
 - 2. Well screen
 - 3. Method of disinfection
 - 4. Method of grouting
 - 5. Method of well development
 - 6. Laboratory to perform chemical analysis
 - 7. Type of drilling equipment

PART 2 MATERIALS

2.1 PERMANENT STEEL CASING PIPE:

A. Steel casing shall be used as a permanent part of the well structure. Steel pipe shall conform to Federal Specifications WW-P-406d, Weight A, Class I, AWWA Standard A-100 and ASTM or API specifications for water well construction. Pipe shall be of standard brand manufacture and have the brand name clearly printed on each length of pipe. Manufacturer rejected pipe shall not be used. Casing shall be free of oil, grease, rust or other detrimental materials

B. Steel casing shall meet the following requirements:

Nominal Diameter (inches)	Nominal Wall Thickness (inches)
6	0.280
8	0.322
10	0.365
12	0.375

C. Lengths of casing shall be connected with either screwed couplings or approved welds. Welded joints shall be made in accordance with the provisions of Section 3.3.

2.2 GROUTING MATERIAL

A. The grout shall be proportioned of Portland Cement and the least quantity of water (not over six gallons per cubic foot of cement) required to give a mixture of such consistency that it can be forced through the grout pipes. Three percent to five percent (by weight) of bentonite clay may be added to the grout mixture in order to improve its flow properties. The mixture, method of mixing, and consistency shall be approved by the Contracting Officer before grouting of the wells

PART 3 WORKMANSHIP

3.1 GENERAL REQUIREMENTS

A. Groundwater well construction shall be in accordance with IDAPA 37.03.09 – Well Construction Standards Rules and other pertinent standards.

B. Construction waste water or stormwater runoff from the construction site shall not contain any sediment, well drilling chemicals or sealant materials.

C. Contractor shall dispose of all well construction waste and debris off site in a legal manner. On-site disposal of any waste materials will not be permitted.

- D. Discharge water from well drilling and test pumping shall be directed away from building and property lines.
- E. All holes, pits, equipment and chemicals shall be safely stored and/or fenced as per OSHA standards.
- F. The construction site shall be restored as nearly as possible to its original condition as soon as the construction of the well and the test pumping has been completed.

3.2 WELL DRILLING

A. Drilling Requirements

1. The total depth of the well is estimated at 250 feet. The total drill depth at one well site shall not exceed 300 feet.
2. The drilling shall be accomplished by air rotary drilling equipment. The equipment shall be in good repair and be capable of drilling, as specified herein, up to a depth of 300 feet. When sufficient yield is indicated from a water bearing stratum, the Contractor shall, after obtaining the Contracting Officer's approval, develop the well according to the provisions of Section 3.2.E of these Special Provisions and IDAPA 37.03.09.
3. All drilling equipment placed down the hole shall be thoroughly disinfected with 500 ppm chlorine solution as per Section 3.11 of these Special Provisions prior to each use.
4. The Contractor shall take such precautions as are necessary or as may be required to prevent contaminated water or water having undesirable physical or chemical characteristics from entering the well. The Contractor shall also take precautions during construction to prevent contamination or pollution from the ground surface.

B. Sampling of Formation

1. The Contractor shall secure representative samples of all materials encountered in the formations drilled in the well. These samples shall be taken at least as often as required by the following:
 - One sample at the beginning of each change in material encountered.
 - One sample for at least each ten feet in non-water bearing formations or in highly impermeable material one sample every five feet in water producing zones.
2. Each sample shall be at least one quart in size and shall be placed in cloth bags or plastic containers. Each sample container shall be clearly labeled with the location of the well, the depth interval represented by the sample, and the date and time taken. The samples shall be furnished to the Contracting Officer as drilling progresses.

C. Well Yield Tests

1. Each time that a water bearing zone is encountered, the Contractor shall conduct a yield test to determine potential production at each well depth. The yield test shall be conducted by pumping or a suitable air lift method for a minimum period of 15 minutes to determine the approximate water production of the well based on water entering through the open hole.

D. Well Logs and Records

1. Well Logs and Records: The Contractor shall keep an accurate and detailed log of the well, recording the following information:
 - Location of well by ¼ section, range and township.
 - Depth, thickness, type, general characteristics, and drilling characteristics of each material encountered.
 - The depth of the static water level (SWL) and observable changes in SWL with well depth.
 - The total depth of completed well.
 - Size, type and length of the well casing installed.
 - Type, size, number and locations of perforations (or well screen slot size and length of screen).
 - Depth of well seal and quantity of cement installed for the seal.
 - Method and date of bacteriological sampling.
2. Each log shall be kept carefully and accurately on a standard well driller's log form with entries made in sequence so that each can be correlated with the depth of hole at the time. It shall be brought up to date at the end of each working day and shall be furnished to the Contracting Officer within one week after the performance of all work on the well.
3. The Contractor shall complete and file a well log report with Idaho Department of Water Resources on the required report form. The Contractor shall also fill out the "Notice of Completion" form as required by the Idaho Department of Water Resources for purposes of appropriating ground water. The Owner shall be responsible for the proper filing of this document with the County Clerk and Recorder and the Department of Water Resources.
4. The Contractor shall, upon the completion of each well, collect and submit a bacteriological sample to the state health department laboratory for analysis. If a positive sample is analyzed the Contractor will be required to chlorinate the well until a negative sample is obtained and analyzed.
5. The Contractor shall, upon the completion of each well that has a satisfactory bacteriological test, shall collect and submit water quality samples for testing.

E. Well Development

1. The Contractor shall develop the well by such methods as will effectively extract from the water bearing formation the maximum practical quantity of drill cuttings, fine sand, silt, or other fine materials. Development shall bring the well to a maximum yield per foot of drawdown, and the water shall be sand free at the maximum pumping rate.
2. One of the following methods of development shall be used: Surge plungers, compressed air, or high velocity jetting. Neither bailing nor pumping shall be considered to be an adequate development method under this contract. The Contracting Officer shall approve the development method used by the Contractor prior to initiation of the work. If compressed air is used, the compressor shall deliver a minimum of 75 cubic feet of air per minute at a minimum pressure of 100 psi. If the compressor used does not meet the requirements specified above, use of compressed air shall not be considered as development.
3. The development work must be done in a manner that does not cause undue settlement of the stratum above the water bearing formation.
4. Development of the well shall be continued until water pumped from the well at the rate determined by the Contracting Officer is clear and free of sand. The water shall be considered sand free when all samples taken at the determined pumping rate contain less than two parts per million (by weight) of sand. Development shall not be considered as test pumping. Checking an aquifer for yield as described in Section 3.8 shall not be considered to be development time.
5. During the course of the development of the well, the Contractor shall conduct a preliminary production test using the "blow-down" method to estimate the production capacity of the well.

F. Plumbness and Alignment

1. The well shall not deviate from the vertical by more than $\frac{3}{4}^{\circ}$.
2. A straightness and alignment test shall be run by the Contractor after the well is cased. This shall consist of running a pipe of 40 foot length and whose outside diameter is between $\frac{1}{2}$ inch and $\frac{3}{4}$ inch smaller than the inside diameter of the casing to the bottom of the well.
3. If the test pipe binds, does not pass through the well casing freely, or is bent as a result of testing procedure, the Contracting Officer shall have the authority to reject the well as drilled and cased, with no payment to the Contractor, until the deficiency is corrected to the Contracting Officer's satisfaction. No payment will be made to the Contractor for this testing procedure or materials.

3.3 CASING

- A. Casings shall be installed in well as shown on the construction drawings to a depth determined during drilling. The length of casing required will depend

on the formation and the overlying materials. The top of any casing which is cut by torch shall have all rough surfaces ground smooth.

- B. The method of installation shall be at the option of the drilling contractor provided the installation meets the requirements for plumbness and alignment and that the installation process does not alter the shape, size, configuration or strength of the casing. The Contractor shall attach centering guides to the casing and screen to insure installation in the center of the drilled hole. Each section of casing shall be securely coupled or shall be welded to adjacent sections with full penetration circumferential welds in conformance Section 3.3.E of these Special Provisions.
- C. A 10-inch diameter outer steel casing shall be installed from ground surface to at least 38 feet into the ground.
- D. A 6-inch Sch. 40 permanent steel casing shall extend from 2 feet above ground surface to the depth needed to facilitate drilling in fractured rock. Minimum depth is 150 feet.
- E. A 4-inch Sch. 40 permanent PVC casing shall extend from 2 feet above ground surface to the bottom of the total well depth.
- F. Welding
 - 1. Surfaces to be welded shall be free from local scale, rust, grease, paint and other foreign material, except for mill scale which withstands vigorous wire brushing. Ends to be welded shall be factory cut and beveled.
 - 2. All welds shall penetrate the full thickness of the casing wall.
 - 3. The technique of welding employed, the appearance and quality of the welds made, and the methods of correcting defective work shall conform to normally accepted standards of workmanship.
 - 4. No welding shall be done when the temperature of the base metal is lower than 0° Fahrenheit. At temperatures between 0° and 32° Fahrenheit, the surface of all areas within three inches of the edges to be welded shall be heated until they are too hot to touch before welding is started.
 - 5. Sections of casing after welding shall be true to line and free from twists, bends, offsets, and openings. All welded joints shall be watertight.

3.4 TEMPORARY WELL CAP

- A. At all times during the progress of the work, the Contractor shall protect the well in such manner as effectively to prevent either tampering with the well or the entrance of foreign matter into it. During construction, the casing shall be temporarily terminated approximately 24 inches above finished ground surface to prevent entry of contaminants. The ground surface around the well shall be graded so that surface water flows away from the well. Borrow

material and topsoil shall be provided as necessary to grade the site to drain away from the well.

- B. The casing shall be cut squarely and uniform with rough edges ground off and suitable for installation of the temporary well cap.

3.5 WELL SCREEN

- A. The Contractor shall furnish and install a 4-inch Sch. 40 PVC well screen with 0.050 Slots. The well screen shall have a tail pipe a minimum of 10 feet in length.
- B. Gravel pack installation should occur in the field unless the formation is stable, hard rock.
- C. Saw cut perforations are acceptable in hard, fractured rock. Saw cut perforations shall be 1/8-inch wide. The total lengths of slots shall be at least 12 feet.

3.6 GROUTING

- A. The well shall have the annular space between the permanent casing and the drill hole filled with cement grout as shown on the construction drawings.
- B. Where a bore hole larger than the outside diameter of the casing has been drilled below the depth to which the casing is to be grouted, the lower part of the hole shall either be backfilled with sand, cement, grout or other material approved by the Contracting Officer. No payment will be made for cement grout or backfill used for this purpose.
- C. At the direction of the Contracting Officer the depth of grouting may be increased where necessary to provide a continuous seal between aquifers. If required, this work shall be completed at the unit price for grouting. No additional compensation beyond the unit price will be made for additional grout to provide a formation seal.
- D. Before proceeding with placing the grout the Contractor shall secure the Contracting Officer's approval of the method he proposes to use. No method will be approved that does not specify the forcing of the grout under pressure from the bottom of the space to be grouted towards the surface. The grouting shall be done continuously and in a manner that will insure the filling of the entire annular space with grout in a single operation.
- E. The volume of grout placed shall be recorded and checked against the volume calculations provided by the Contracting Officer every 10 feet of well depth being grouted. Top of seal will be tagged (sounded) by Contracting Officer every 10 feet as grout is placed to compare actual volume placed against volume calculations.

- F. The Contractor shall provide a minimum of 50% more grout material than the volume calculations, provided by the Contracting Officer, indicate will be required.
- G. The casing or tremie pipe used to place the grout shall be withdrawn as the grout is placed.
- H. Contractor shall notify Contracting Officer a minimum of 24 hours prior to placement of grout seal. Contracting Office must be present when grout seal is placed.
- I. Contractor shall notify Idaho Water Resources (IDWR) a minimum of 24 hours prior to placement of grout seal.
- J. If separate aquifers of unequal quality or pressure are encountered during drilling operations, intermediate seals will be required.
- K. No drilling, test pumping or other work in the well will be permitted within 72 hours after completion of grouting. If quick set cement is used, this period may be reduced to 24 hours, upon approval by the Contracting Officer.

3.7 TESTING FOR YIELD AND DRAWDOWN

- A. The Contractor shall furnish all labor, equipment materials and power required for test pumping, to the satisfaction of the Contracting Officer. This shall include a calibrated water meter to measure the rate of pump discharge. The Contractor shall also provide and install a 1" diameter PVC tube in the well for the insertion of electrical probes to measure the water level during the drawdown test. An alternative method for method of measuring water level may be approved if proposed by the well driller.
- B. The Contractor shall furnish and install the required pumping equipment and piping facilities that will be removed from the well at the end of the test pumping operation. The Contractor will also be responsible to provide a generator and the required electrical wiring to operate the temporary pumping equipment during the test pumping operation. The well shall be test pumped at a rate of 100% of the estimated well production capacity for 24 hours or at a rate of 150% of the estimated capacity for 6 hours unless otherwise directed by the Contracting Officer.
- C. The duration of the test pumping shall not be less than twenty-four (24) hours at 100% of the estimated well production capacity or less than six (6) hours at 150% of the estimated capacity after the water level has stabilized in the well. If flow is estimated to be between 5 gpm and 10 gpm, test duration shall be extended to seventy-two (72) hours. The Contracting Officer may require more or less test pumping time. Before completion of the test pumping, the Contractor shall collect the required water samples and send them to the laboratory for the required chemical analysis as per Section 3.10.

3.8 CHEMICAL ANALYSIS

A. The Contractor shall furnish to the Contracting Officer a chemical analysis of the water produced from the well. The Contractor shall be responsible for furnishing the required size and number of containers, collecting the water sample, arranging for the chemical analysis to be performed at a laboratory approved by the Contracting Officer and furnishing the results to the Contracting Officer. The following is a list of the parameters to be analyzed for the well:

1. Total Coliform/E. Coli
 - a. Coliform Presence/Absence
 - b. E. Coli Enumeration
2. Inorganic contaminants (IOC's)
 - a. Idaho Complete IOC (Phase V, II, and optional)
 - i. Corrosivity
3. Organic contaminants (VOC's/SOC's)
 - a. Complete Phase II and V SOC/VOC
4. Radionuclides
 - a. Gross Alpha/Beta Radioactivity
 - b. Radium 228
 - c. Radium 226
 - d. Uranium

3.9 DISINFECTION AND BACTERIOLOGICAL TEST

- A. All down-hole materials and equipment used for drilling and test pumping shall be cleaned and disinfected with a 500 ppm chlorine solution as per IDWR rules.
- B. Upon completion of construction and test pumping, the well shall be thoroughly cleaned of foreign substances including but not limited to cement, oil, grease, joint dope, and scum. The well shall then be disinfected with a chlorine solution. The Contractor shall obtain approval from the Contracting Officer of his disinfection procedures based on his submittals.
- C. The chlorine solution used for disinfecting the well shall be prepared prior to injection into the well and shall be of such volume and strength and so applied that a minimum concentration of 50 ppm of free residual chlorine shall be obtained in all parts of the well for at least 24 hours after injection. The portion of the casing above the static water level shall be flushed with the chlorine solution.

D. The chlorine may be added to the well by the following method:

1. A stock solution of chlorine will be added through a pipe discharging at different levels in the well. The well will then be agitated with a bit or bailer to mix the solution.

E. Following the final disinfection of the well, the Contractor shall flush the well to remove all of the chlorine from the well.

F. Following the flushing of the well, the Contractor shall be responsible for collecting a water sample from the well and having it tested for total coliform and E. Coli bacteria at a testing laboratory approved by the Contracting Officer. If the water sample does not pass the bacteriological test, then the Contractor will be required to disinfect the well again and then retest for total coliform and E. Coli again. This process will be repeated until the water sample passes the bacteriological test. The Contractor will be paid for one (1) bacteriological test. No additional payment will be made to the Contractor for additional bacteriological tests required.

3.13 ABANDONMENT OF WELL

A. In the event the well shall not be accepted for completion due to insufficient capacity; unsatisfactory chemical or bacteriological quality; or should it be abandoned because of poor alignment, loss of tools, or for any other cause, the Contractor shall as directed by the Contracting Officer, fill the abandoned hole with puddle clay or clay and concrete. No well shall be abandoned without the approval of the Contracting Officer. Provision for payment for drilling and materials used for abandoned well shall be made as follows:

1. In cases where abandonment of the well is due to causes which are determined not to be the fault of the Contractor, casing and other materials furnished by the Contractor but not installed shall remain his property and no payment will be made for such materials. The Contractor shall be paid only the unit cost for drilling and the unit cost for materials installed prior to abandonment of the well.
2. Any well hole abandoned due to the fault of the Contractor before completion and acceptance shall be the responsibility of the Contractor and no payment shall be made for such work.

PART 4 MEASUREMENT AND PAYMENT

- 4.1 Use the following unit prices. Includes all labor, tools, equipment and materials, and performing all operations necessary to drill, case, develop, and the groundwater well as specified.

Due to the nature of well construction, actual quantities may vary substantially from bid quantities. No allowance will be made in unit prices for substantial variations in quantities.

- A. Drilling Permit: Per each. Includes all labor, materials and incidentals required to obtain the drilling permit.
1. Bid Schedule Payment Reference: SP 2100.4.1.A.
 2. Bid Schedule Description: Drilling Permit...each (EA).
- B. 12" Drilling: By the linear foot vertically from the existing ground surface to the bottom of the 12" diameter drilled hole. Includes all materials, labor, equipment and incidentals required to complete the work as specified regardless of materials encountered during the drilling. No additional payment will be made for drilling in rock, including boulders. It shall be the Contractor's responsibility to familiarize themselves with the particular area and bid accordingly.
1. Bid Schedule Payment Reference: SP 2100.4.1.B.
 2. Bid Schedule Description: 12" Drilling...linear foot (LF).
- C. 10" Drilling: By the linear foot vertically from the bottom of the 12" diameter drilled hole to the bottom of the 10" diameter drilled hole. Includes all materials, labor, equipment and incidentals required to complete the work as specified regardless of materials encountered during the drilling. No additional payment will be made for drilling in rock, including boulders. It shall be the Contractor's responsibility to familiarize themselves with the particular area and bid accordingly.
1. Bid Schedule Payment Reference: SP 2100.4.1.C.
 2. Bid Schedule Description: 10" Drilling...linear foot (LF).
- D. 10" Diameter Outer Steel Casing: By the linear foot vertically from the ground surface to the bottom of the 12" diameter drilled hole. Includes all labor, equipment and incidentals required to install the 10" temporary steel casing as specified herein.
1. Bid Schedule Payment Reference: SP 2100.4.1.D.
 2. Bid Schedule Description: 10" Diameter Outer Steel Casing...linear foot (LF).
- E. 6" Diameter Steel Well Casing: By the linear foot vertically from a point 2 feet above the ground surface to a minimum depth of 150 feet to facilitate drilling in fractured rock. Includes all materials, labor, equipment and incidentals required to install the 6" diameter steel well casing as specified.
1. Bid Schedule Payment Reference: SP 2100.4.1.E.
 2. Bid Schedule Description: 6" Diameter Steel Well Casing ...linear foot (LF).
- F. 4" Diameter PVC Well Casing: By the linear foot vertically from a point 2 feet above the ground surface to the bottom of the 10" diameter drilled hole. Includes all materials, labor, equipment and incidentals required to install the 4" diameter PVC well casing as specified.

1. Bid Schedule Payment Reference: SP 2100.4.1.F.
 2. Bid Schedule Description: 4" Diameter PVC Well Casing ...linear foot (LF).
- G. PVC Well Screen: By the linear foot vertically from the bottom of the well screen to the top of the well screen. Includes all materials, labor, equipment and incidentals required to install the well screen in accordance with the specifications and as shown on the plans.
1. Bid Schedule Payment Reference: SP 2100.4.1.G.
 2. Bid Schedule Description: Continuous Slot Well Screen ...linear foot (LF).
- H. 6" Temporary Well Cap: Per each. Includes all materials, labor, equipment and incidentals required to install the 6" temporary well cap as specified.
1. Bid Schedule Payment Reference: SP 2100.4.1.H.
 2. Bid Schedule Description: 6" Temporary Well Cap ... each (EA).
- I. Grouting: By the linear foot vertically from the ground surface to the bottom of the 12" diameter drilled hole. Includes all materials, labor, equipment and incidentals required to install the grout as specified.
1. Bid Schedule Payment Reference: SP 2100.4.1.I.
 2. Bid Schedule Description: Grouting... linear foot (LF).
- J. Testing for Yield and Drawdown: By the hour. Includes all materials, labor, equipment and incidentals required to complete the yield and draw-down test as specified.
1. Bid Schedule Payment Reference: SP 2100.4.1.J
 2. Bid Schedule Description: Testing for Yield and Drawdown... each (EA).
- K. Chemical Analysis of Water: By the lump sum. Includes all materials, labor, equipment and incidentals required to complete the chemical analysis as specified.
1. Bid Schedule Payment Reference: SP 2100.4.1.K.
 2. Bid Schedule Description: Chemical Analysis of Water...lump sum (LS).
- L. Bacteriological Test: Per each. Includes all materials, labor, equipment, lab costs and incidentals required to complete the bacteriological testing as specified.
1. Bid Schedule Payment Reference: SP 2100.4.1.L.
 2. Bid Schedule Description: Bacteriological Test...each (EA).

- M. “Priority 2” Well Drilling & Development: By the linear foot vertically from the ground surface to the bottom of the well. Includes all materials, labor, equipment, lab costs and incidentals required to drill, develop and test the “Priority 2” well location, if necessary following items A through L as specified above.
1. Bid Schedule Payment Reference: SP 2100.4.1.M.
 2. Bid Schedule Description: “Priority 2” Well Drilling & Development...linear foot (LF).
- N. Abandon “Priority 1” / “Priority 2” Well: By the linear foot vertically from the ground surface to the bottom of the well. Includes all materials, labor, equipment, necessary to decommission the ground water well(s), if necessary, per IDAPA 37.03.09.
1. Bid Schedule Payment Reference: SP 2100.4.1.N.
 2. Bid Schedule Description: Abandon Well...linear foot (LF).

END OF SECTION